



NeuroSpace

Complete evaluation export

Job ID	e579c7d3-8398-4621-9f67-c82570b4662b
Status	completed
Result timestamp	2026-05-13T19:41:59.409480
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Pipeline Configuration

Parameter	Value
model_types	rnn, qrnn
comparison_mode	mi
feature_selector	mi
manual_overrides.rnn_epochs	30
manual_overrides.rnn_batch_size	64
manual_overrides.qrnn_epochs	15
manual_overrides.qrnn_batch_size	32
num_classes	6
class_names	1, 2, 3, 4, 5, 6
qrnn_backend	pennylane
noise_enabled	True
noise_level	0.0100
mitigation_enabled	True
mitigation_runs	3

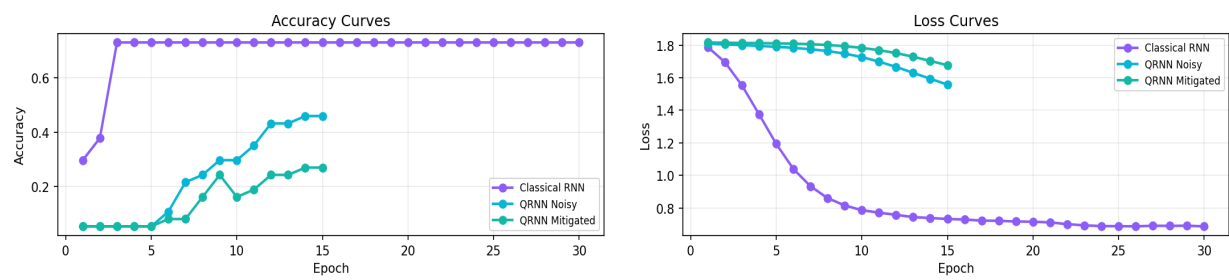
Best Model

Winner	RNN Accuracy	QRNN Accuracy	Improvement
RNN	81.08%	41.89%	39.19%

Evaluation Summary

Model	Accuracy	Precision	Recall	F1 Score	Loss
Classical RNN	81.08%	81.14%	81.08%	80.05%	0.6877
QRNN Noisy	41.89%	55.13%	41.89%	42.05%	1.5578
QRNN Mitigated	39.19%	48.35%	39.19%	36.36%	1.6766

Learning Curves



Classical RNN

Metric	Value
Accuracy	81.08%
Precision	81.14%
Recall	81.08%
F1 Score	80.05%
Loss	0.6877
Train Loss	0.6061
Val Loss	0.6877
Train Accuracy	77.65%
Val Accuracy	72.97%
Train Size	255
Val Size	37
Test Size	74

Classical RNN Confusion Matrix

Actual / Predicted	Class 0	Class 1				
Class 0	23	0	0	0	0	0
Class 1	0	3	0	5	1	3
Class 2	0	0	15	0	0	0
Class 3	0	2	0	6	0	2
Class 4	0	0	0	0	10	0
Class 5	0	1	0	0	0	3

Classical RNN Classification Report

Class	Precision	Recall	F1 Score	Support
1	100.00%	100.00%	100.00%	23.0000
macro avg	72.16%	76.67%	72.62%	74.0000
weighted avg	81.14%	81.08%	80.05%	74.0000
Accuracy			81.08%	

Classical RNN Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	1.9252	1.7883	20.78%	29.73%
2	1.4951	1.6956	47.06%	37.84%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
3	1.1762	1.5525	72.16%	72.97%
4	1.0192	1.3749	74.90%	72.97%
5	0.9411	1.1940	76.08%	72.97%
6	0.8524	1.0406	80.00%	72.97%
7	0.8320	0.9341	76.08%	72.97%
8	0.7954	0.8614	75.69%	72.97%
9	0.7452	0.8162	80.39%	72.97%
10	0.7065	0.7873	79.22%	72.97%
11	0.7225	0.7718	79.22%	72.97%
12	0.7205	0.7587	77.65%	72.97%
13	0.6654	0.7445	80.78%	72.97%
14	0.7048	0.7394	81.18%	72.97%
15	0.6604	0.7332	79.61%	72.97%
16	0.6275	0.7300	80.39%	72.97%
17	0.6578	0.7230	80.78%	72.97%
18	0.6354	0.7222	80.78%	72.97%
19	0.6466	0.7178	78.43%	72.97%
20	0.6399	0.7161	80.39%	72.97%
21	0.6235	0.7124	80.00%	72.97%
22	0.6570	0.7015	78.04%	72.97%
23	0.6174	0.6943	79.61%	72.97%
24	0.6282	0.6896	77.25%	72.97%
25	0.6226	0.6891	77.65%	72.97%
26	0.6239	0.6880	78.82%	72.97%
27	0.5959	0.6916	78.82%	72.97%
28	0.6008	0.6909	81.18%	72.97%
29	0.5942	0.6919	80.39%	72.97%
30	0.6061	0.6877	77.65%	72.97%

QRNN Noisy

Metric	Value
Accuracy	41.89%
Precision	55.13%
Recall	41.89%
F1 Score	42.05%
Loss	1.5578
Train Loss	1.5445
Val Loss	1.5578
Train Accuracy	43.53%
Val Accuracy	45.95%
Train Size	255
Val Size	37
Test Size	74

QRNN Noisy Confusion Matrix

Actual / Predicted	Class 0	Class 1				
Class 0	12	4	4	2	0	1
Class 1	0	0	1	8	0	3
Class 2	3	0	8	2	0	2
Class 3	0	0	0	7	0	3
Class 4	0	0	0	7	2	1
Class 5	0	0	0	2	0	2

QRNN Noisy Classification Report

Class	Precision	Recall	F1 Score	Support
1	80.00%	52.17%	63.16%	23.0000
macro avg	47.20%	40.92%	35.91%	74.0000
weighted avg	55.13%	41.89%	42.05%	74.0000
Accuracy			41.89%	

QRNN Noisy Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	1.8036	1.8088	5.49%	5.41%
2	1.7950	1.8044	5.49%	5.41%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
3	1.7880	1.7996	5.49%	5.41%
4	1.7810	1.7948	5.49%	5.41%
5	1.7744	1.7896	5.49%	5.41%
6	1.7658	1.7829	7.84%	10.81%
7	1.7548	1.7747	22.75%	21.62%
8	1.7419	1.7635	38.43%	24.32%
9	1.7263	1.7477	44.71%	29.73%
10	1.7058	1.7265	44.71%	29.73%
11	1.6791	1.6987	45.10%	35.14%
12	1.6542	1.6661	44.31%	43.24%
13	1.6192	1.6300	43.53%	43.24%
14	1.5855	1.5933	42.35%	45.95%
15	1.5445	1.5578	43.53%	45.95%

QRNN Mitigated

Metric	Value
Accuracy	39.19%
Precision	48.35%
Recall	39.19%
F1 Score	36.36%
Loss	1.6766
Train Loss	1.6183
Val Loss	1.6766
Train Accuracy	37.65%
Val Accuracy	27.03%
Train Size	255
Val Size	37
Test Size	74

QRNN Mitigated Confusion Matrix

Actual / Predicted	Class 0	Class 1				
Class 0	4	5	1	0	6	7
Class 1	0	0	1	11	0	0
Class 2	2	0	8	3	1	1
Class 3	0	0	0	10	0	0
Class 4	0	0	0	1	7	2
Class 5	0	0	0	4	0	0

QRNN Mitigated Classification Report

Class	Precision	Recall	F1 Score	Support
1	66.67%	17.39%	27.59%	23.0000
macro avg	38.52%	40.12%	33.53%	74.0000
weighted avg	48.35%	39.19%	36.36%	74.0000
Accuracy			39.19%	

QRNN Mitigated Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	1.8171	1.8161	5.49%	5.41%
2	1.8065	1.8148	5.49%	5.41%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
3	1.8001	1.8131	5.49%	5.41%
4	1.7931	1.8124	5.49%	5.41%
5	1.7847	1.8107	8.24%	5.41%
6	1.7774	1.8091	19.22%	8.11%
7	1.7661	1.8055	20.39%	8.11%
8	1.7575	1.8008	20.39%	16.22%
9	1.7453	1.7931	20.39%	24.32%
10	1.7328	1.7829	24.31%	16.22%
11	1.7136	1.7690	35.29%	18.92%
12	1.6959	1.7513	35.69%	24.32%
13	1.6708	1.7283	36.47%	24.32%
14	1.6468	1.7028	36.86%	27.03%
15	1.6183	1.6766	37.65%	27.03%